

Program Question_3

Implicit None

Integer::i

Write(*,*)"Fibonacci number Using Iterative function"

Do i=0,19

Write(*,*)" (Fib",i,")= ",fib_iter(i)

End Do

Write(*,*)" "

Write(*,*)"Fibonacci number using Recursive Function"

Do i=0,19

Write(*,*)" (Fib",i,")= ",fib_recursive(i)

End Do

Contains

!Iterative function to calculate Fibonacci number

Function fib_iter(n) result(fib)

Integer,Intent(In)::n

Integer::a,b,fib,i,temp

If(n==0)Then

fib=0

else if(n==1)Then

fib=1

else

a=0

b=1

Do i=2,n

temp=a+b

a=b

b=temp

End Do

fib=b

End If

End Function fib_iter

!Recursive Function to calculate Fibonacci Number

Recursive Function fib_recursive(n) Result(fib)

Integer,Intent(In)::n

Integer::fib

If(n==0)then

fib=0

else if(n==1)then

fib=1

else

fib=fib_recursive(n-1)+fib_recursive(n-2)

End If

End Function

End Program